Problem 3

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Problem 1 *Assume that the rate of change (in dollars per day) of the price of shares of stock in the WeSaySo Company (with t in days) is modeled by the equation $r(t) = -3t^2 + 30t - 63$ (note that this is technically a discrete function, but prices change so often with stocks that modeling this with a continuous function makes sense). Assume also that the price of a share of stock on day 1 (i.e., $t = 1$) is \$51. Answer the following questions:*

(a) *Find the rate of change of price at $t = 5$.*

Explanation. $r(5) = -3(25)+30(5)-63 = -75+150-63 = 12$ dollars/day.

(b) *Find the price of a share of stock at $t = 5$.*

Explanation. Let $p(t)$ denote the price of a share of stock at any time t. Then notice that $r(t) = p'(t)$. We will first find $p(t)$ for general t, and then substitute $t = 5$ to solve this problem.

$$\begin{aligned} p(t) &= \int_1^t r(s)\,ds + p(1) \\ &= \int_1^t \left(-3s^2 + 30s - 63\right)\,ds + 51 \\ &= \Big[-s^3 + 15s^2 - 63s\Big]_1^t + 51 \\ &= \left(-t^3 + 15t^2 - 63t\right) - (-1 + 15 - 63) + 51 \\ &= -t^3 + 15t^2 - 63t + 100. \end{aligned}$$

So, $p(5) = -125 + 375 - 315 + 100 = 35$\$.

(c) *How fast is the rate of change of price changing at $t = 5$?*

Explanation. $r'(t) = -6t + 30$. So, $r'(5) = -30 + 30 = 0$.

(d) *How much did the price of a share of stock change in the first 6 days (i.e., on $[1, 6]$)?*

Explanation. $p(6) - p(1) = (-216 + 540 - 378 + 100) - 51 = -5$\$.

(e) *What was the greatest rate of change of price during the first 6 days (i.e., on $[1, 6]$)? [3]*

Explanation. This question wants us to maximize $r(t)$ on the closed interval $[1,6]$. So we need to find all critical points of $r(t)$ on $[1,6]$.

$$r'(t) = -6t + 30 := 0 \quad \Longrightarrow \quad t = 5$$

Then, using the closed interval method, we simply plug $t = 1, 5, 6$ into $r(t)$ and check which has the greatest output.

$$\begin{aligned} r(1) &= -3 + 30 - 63 = -36 \\ r(5) &= -75 + 150 - 63 = 12 \\ r(6) &= -108 + 180 - 63 = 9. \end{aligned}$$

Thus, the greatest rate of change of price is 12 dollars/day when $t = 5$.

(f) *What was the greatest price of a share of stock during the first* 6 *days (i.e., on* $[1,6]$*)? [3]*

Explanation. This question wants us to maximize $p(t)$ on the closed interval $[1,6]$. So we need to find all critical points of $p(t)$ on $[1,6]$. But note that $p'(t) = r(t)$. So critical points of $p(t)$ are just roots of $r(t)$.

$$r(t) = 0$$

$$-3t^2 + 30t - 63 = 0$$

$$-3(t^2 - 10t + 21) = 0$$

$$-3(t-3)(t-7) = 0$$

$$t = 3, 7 \quad \Longrightarrow \quad t = 3$$

since 7 is not in the interval $[1,6]$. So we use the Closed Interval Method:

$$\begin{aligned} p(1) &= -1 + 15 - 63 + 100 = 51 \\ p(3) &= -27 + 135 - 189 + 100 = 19 \\ p(6) &= -216 + 540 - 378 + 100 = 46 \end{aligned}$$

Thus, the greatest price of the stock is \$51 when $t = 1$.